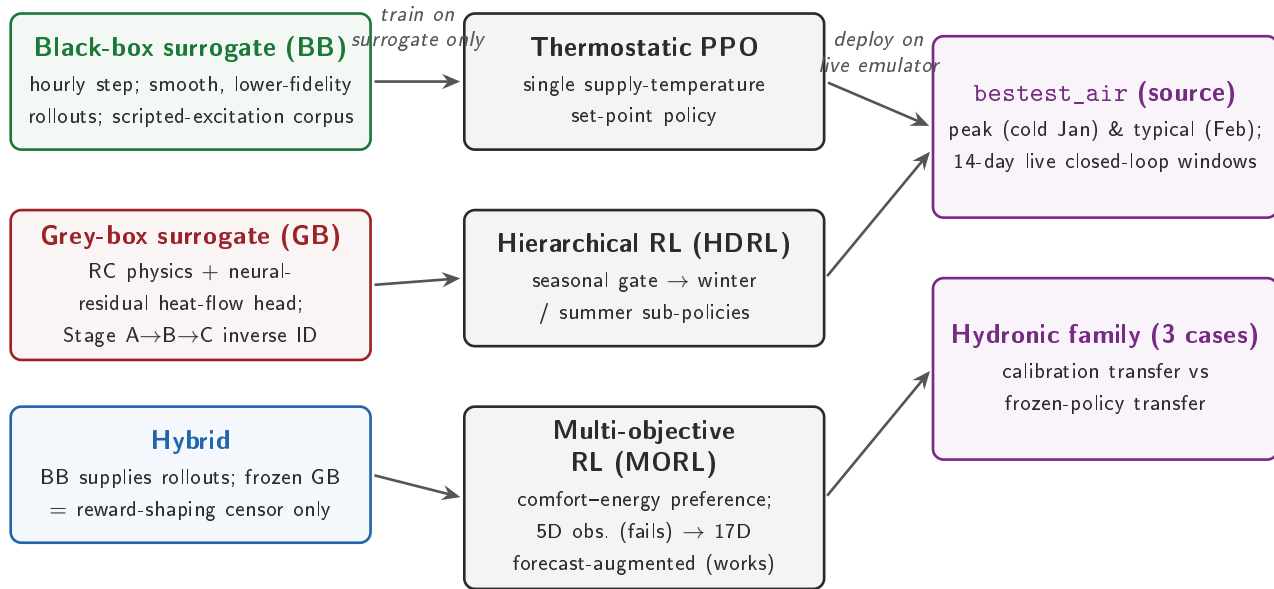


Surrogate training environments

Controller families (all PPO-based)

Live BOPTEST evaluation



Live metric: maintenance score m_s = time-in-violation + worst-case severity
($m_s > 1$ = unusable); also comfort-violation %, energy, and 24 h rollout RMSE_T .